

Classification of suicide attempters in schizophrenia using sociocultural and clinical features: A machine learning approach



Nuwan C. Hettige^{a,b}, Thai Binh Nguyen^a, Chen Yuan^a, Thanara Rajakulendran^a, Jermeen Baddour^a, Nikhil Bhagwat^g, Ali Bani-Fatemi^{a,b}, Aristotle N. Voineskos^{b,c}, M. Mallar Chakravarty^{d,e,f}, Vincenzo De Luca^{a,b,c,*}

^a Group for Suicide Studies, Centre for Addiction and Mental Health, 250 College Street, Toronto, Ontario M5T 1R8, Canada

^b Institute of Medical Science, University of Toronto, Toronto, Ontario, Canada

^c Department of Psychiatry, University of Toronto, Toronto, Ontario, Canada

^d Douglas Mental Health University Institute, McGill University, Montreal, Canada

^e Department of Psychiatry, McGill University, Montreal, Canada

^f Biological and Biomedical Engineering, McGill University, Montreal, Canada

^g Institute of Biomaterials & Biomedical Engineering, University of Toronto, Toronto, Canada

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ABSTRACT

Objective: Suicide is a major concern for those afflicted by schizophrenia. Identifying patients at the highest risk for future suicide attempts remains a complex problem for psychiatric interventions. Machine learning models allow for the integration of many risk factors in order to build an algorithm that predicts which patients are likely to attempt suicide. Currently it is unclear how to integrate previously identified risk factors into a clinically relevant predictive tool to estimate the probability of a patient with schizophrenia for attempting suicide.

Methods: We conducted a cross-sectional assessment on a sample of 345 participants diagnosed with schizophrenia spectrum disorders. Suicide attempters and non-attempters were clearly identified using the Columbia Suicide Severity Rating Scale (C-SSRS) and the Beck Suicide Ideation Scale (BSS). We developed four classification algorithms using a regularized regression, random forest, elastic net and support vector machine models with sociocultural and clinical variables as features to train the models.

Results: All classification models performed similarly in identifying suicide attempters and non-attempters. Our regularized logistic regression model demonstrated an accuracy of 67% and an area under the curve (AUC) of 0.71, while the random forest model demonstrated 66% accuracy and an AUC of 0.67. Support vector classifier (SVC) model demonstrated an accuracy of 67% and an AUC of 0.70, and the elastic net model demonstrated and accuracy of 65% and an AUC of 0.71.

Conclusion: Machine learning algorithms offer a relatively successful method for incorporating many clinical features to predict individuals at risk for future suicide attempts. Increased performance of these models using clinically relevant variables offers the potential to facilitate early treatment and intervention to prevent future suicide attempts.

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1. Introduction

Schizophrenia is a mental disorder which is associated with higher incidences of suicidal behaviour [11,25] which remains the leading cause of death among psychiatric patients.

There appears to be a significant increase in the risk for suicide among those with schizophrenia as compared to the general population [6]. Previous epidemiological studies have conservatively suggested a

4.9% risk for suicide among this population [25] and approximately 20–40% attempting suicide during their lifetime [28]. While many patients may attempt suicide due to psychosocial stressors associated with the disorder, there are also a significant number of patients with the same stressors who do not attempt suicide. Many risk factors have been proposed to explain why some individuals may be driven to attempt suicide [29] but these mainly delineate the differences between suicide attempters and non-attempters without adding predictive power. In order to improve the predictive accuracy of these risk factors, it is imperative that they be integrated to establish a predictive model that may be able to estimate the likelihood of an individual patient with schizophrenia to attempt suicide.

* Corresponding author at: Centre for Addiction and Mental Health, Department of Psychiatry, University of Toronto, 250 College St, Toronto, ON, M5T 1R8, Canada.
E-mail address: vincenzo_deluca@camh.net (V. De Luca).

Being able to accurately predict or identify those at risk for future suicide attempts is a major challenge in psychiatry and, in particular, among schizophrenia patients. Many previous studies have attempted to identify clinical variables that may help predict the risk for suicide in schizophrenia. However, such efforts have largely been unsuccessful, producing many false positives or inconsistent results between studies [28]. Creating a statistical or multivariate framework that successfully incorporates the role of potential clinical and socio-cultural risk factors may allow for an accurate prediction of suicide attempt risk for an individual patient. Being able to predict suicide risk at the individual level will undoubtedly allow for improved efforts in reducing suicide attempts among high risk patients in this population [10].

The prediction of suicide attempters and non-attempters in schizophrenia is a complex classification problem that requires the simultaneous inclusion of many potential risk factors [32]. Classification is defined as the process of assigning individuals to one or more classes – in this case, suicide attempters or non-attempters. Machine learning algorithms in psychiatric research are emerging as a valuable tool to incorporate many predictors and improve classification capabilities among unseen observations (*i.e.* patients) [26]. Algorithms that include a small number of rules or operations, such as in general linear models, may likely solve simple classification problems in psychiatry. However, complex classification problems, such as suicide attempt status, require algorithms that model complex relationships among a large number of factors [2,33]. Machine learning models are designed to consider these complex relationships to determine an optimal classification model. In fact, recent studies have demonstrated some success in utilizing these models in suicide research among U.S. soldiers [19].

Many individual risk factors have been put forward that may contribute to suicide risk in schizophrenia. To develop our algorithm, we considered sociocultural and clinical risk factors suggested in the literature. Feature (variable) selection for our model was based on *a priori* knowledge of sociocultural risk factors for suicide attempts associated with demographic, illness-related, sociocultural, comorbid, and clinical scale variables. In the current study we implemented four machine learning algorithms: a least absolute shrinkage and selection operator (LASSO) with a general binary logistic model [37], Random Forests

(RF) [5], elastic net and support vector machine (SVM) learning classifier. Our main objective was to establish a clinically useful predictive model able to accurately identify suicide attempters and non-attempters among our sample of schizophrenia patients. Our hypothesis is that the combination of several clinical predictors can increase the accuracy of the model.

2. Materials and methods

2.1. Study sample

We recruited a final sample of 345 participants from the Centre for Addiction and Mental Health (CAMH), a Canadian teaching hospital located in Toronto. The patients were all in follow up at CAMH and they were treated by a staff psychiatrist. Participant inclusion criteria were ages of 18 to 75 and meeting the criteria for schizophrenia spectrum disorders based on the structured clinical interview for DSM-IV (SCID-I/P). Evidence of intellectual disability, a history of brain injury trauma with a loss of consciousness, and a diagnosis of drug induced psychosis were considered as exclusion criteria. Written informed consent was obtained for participation in the study as well as for the release of participants' medical history in order to verify clinical data or obtain missing information. This study was approved by the CAMH Research Ethics Board.

2.2. Clinical assessments

We conducted cross-sectional assessments using a structured interview and self-report questionnaires. The interview incorporated the Structured Clinical Interviews for DSM-IV (SCID-I/P) to confirm the participants' primary diagnosis and assess for additional psychiatric symptoms and comorbid diagnoses (*i.e.* lifetime alcohol abuse/dependence and drug abuse/dependence). In cases where a reliable diagnosis could not be clearly defined, the individual was excluded from the analysis. Participants' suicidal history was assessed using the Beck Suicide Ideation Scale (BSS) [1] and the Columbia Suicide Severity Rating Scale (C-SSRS) [30]. According to the patients' response on the BSS and C-SSRS, they were classified as suicide attempters (those who have

Table 1
Summary statistics for clinical, demographic, and sociocultural predictor variables (mean $\pm$ SD).

Total (N = 345)	Suicide non-attempter (N = 214)	Suicide attempter (N = 131)	P-value	Odds ratio	95% confidence interval
Age (years) (M $\pm$ S)	38.89 $\pm$ 12.92	44.51 $\pm$ 12.62	1.35×10^{-4}	1.034	1.016–1.052
Sex (male/female)	157/57	84/47	0.070	0.649	0.406–1.036
Duration of illness (M $\pm$ S)	15.91 $\pm$ 11.39	22.09 $\pm$ 12.52	7.97×10^{-6}	1.043	1.024–1.063
Number of hospitalizations (N > 5)	50 (23%)	61 (47%)	1.05×10^{-5}	2.858	1.797–4.578
Immigrated before age 9	22 (10%)	19 (15%)	0.241	1.481	0.768–2.855
Immigrated between ages 10–18	31 (15%)	16 (12%)	0.551	0.821	0.430–1.568
Immigrated after age 18	14 (7%)	11 (8%)	0.520	1.310	0.576–2.978
Self-reported ethnicity	108 (55%)	77 (59%)	0.134	1.400	0.902–2.171
Religious identity	155 (72%)	103 (79%)	0.199	0.714	0.427–1.194
English as primary language	183 (86%)	111 (85%)	0.843	1.064	0.578–1.957
Family history of psychosis	57 (27%)	51 (39%)	0.017	1.756	1.104–2.792
Family History of mood disorders	76 (36%)	65 (50%)	0.010	1.788	1.150–2.789
Family history of suicidal behaviour	34 (16%)	47 (36%)	3.19×10^{-5}	2.962	1.782–4.972
NEO neuroticism (M $\pm$ S)	33.47 $\pm$ 8.36	36.75 $\pm$ 8.61	7.14×10^{-4}	1.047	1.020–1.076
NEO extraversion (M $\pm$ S)	37.19 $\pm$ 6.60	37.86 $\pm$ 7.67	0.672	0.993	0.963–1.025
NEO openness (M $\pm$ S)	39.11 $\pm$ 6.66	39.24 $\pm$ 6.46	0.856	1.003	0.970–1.037
NEO agreeableness (M $\pm$ S)	40.34 $\pm$ 5.76	40.40 $\pm$ 5.20	0.958	1.001	0.962–1.041
NEO conscientiousness (M $\pm$ S)	42.45 $\pm$ 7.55	41.41 $\pm$ 7.82	0.221	0.982	0.955–1.011
CTQ physical abuse (M $\pm$ S)	7.51 $\pm$ 3.69	9.66 $\pm$ 5.25	3.39×10^{-5}	1.114	1.060–1.175
CTQ emotional abuse (M $\pm$ S)	9.68 $\pm$ 4.58	12.90 $\pm$ 5.99	2.03×10^{-7}	1.121	1.075–1.171
CTQ sexual abuse (M $\pm$ S)	7.29 $\pm$ 4.09	8.85 $\pm$ 5.39	3.23×10^{-3}	1.072	1.024–1.125
CTQ physical neglect (M $\pm$ S)	8.48 $\pm$ 3.49	9.56 $\pm$ 4.29	0.013	1.075	1.016–1.139
CTQ emotional neglect (M $\pm$ S)	11.07 $\pm$ 4.81	13.24 $\pm$ 5.76	2.54×10^{-4}	1.082	1.038–1.130
CTQ total score (M $\pm$ S)	38.35 $\pm$ 4.74	47.09 $\pm$ 19.20	9.91×10^{-6}	1.031	1.018–1.046
Lifetime alcohol abuse/dependence	63 (29%)	37 (28%)	0.812	0.943	0.583–1.526
Lifetime drug abuse/dependence	62 (29%)	52 (40%)	0.063	1.543	0.978–2.435
Lifetime marijuana abuse/dependence	62 (29%)	31 (24%)	0.282	0.760	0.461–1.252

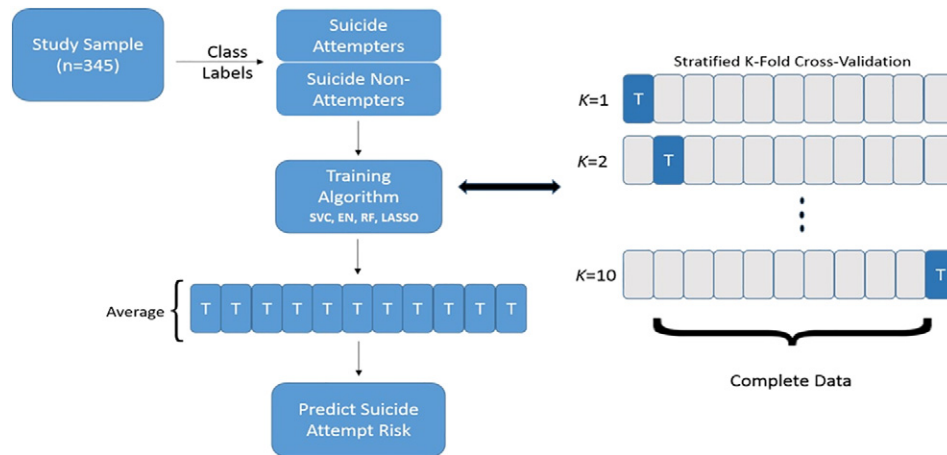


Fig. 1. Machine learning analysis pipeline. Flow chart demonstrating the process by which our machine learning algorithms trained and tested our clinical variables in predicting suicide attempters. Stratified K-fold cross validation was used to maintain class distributions across all folds; T = test data.

attempted suicide at least once in their lifetime) or non-attempters (those who never attempted suicide in their lifetime). Individuals with a contradictory suicide attempt history between self-report and interview-based scales were confirmed using the medical charts and unresolved cases were excluded. Information regarding the participants' ethnicity was collected through self-report and interview questions. Individuals were classified as either European Caucasian or non-European

Caucasian. We also administered the Childhood Trauma Questionnaire (CTQ). The CTQ was used for screening early life adversities because showed an internal consistency of 0.95 [22].

To screen for personality profile we used the NEO Five Factor Inventory (NEO-FFI) scale. The NEO-FFI has shown an internal consistency ranging from 0.73 to 0.87 in a sample of Canadian students [18].

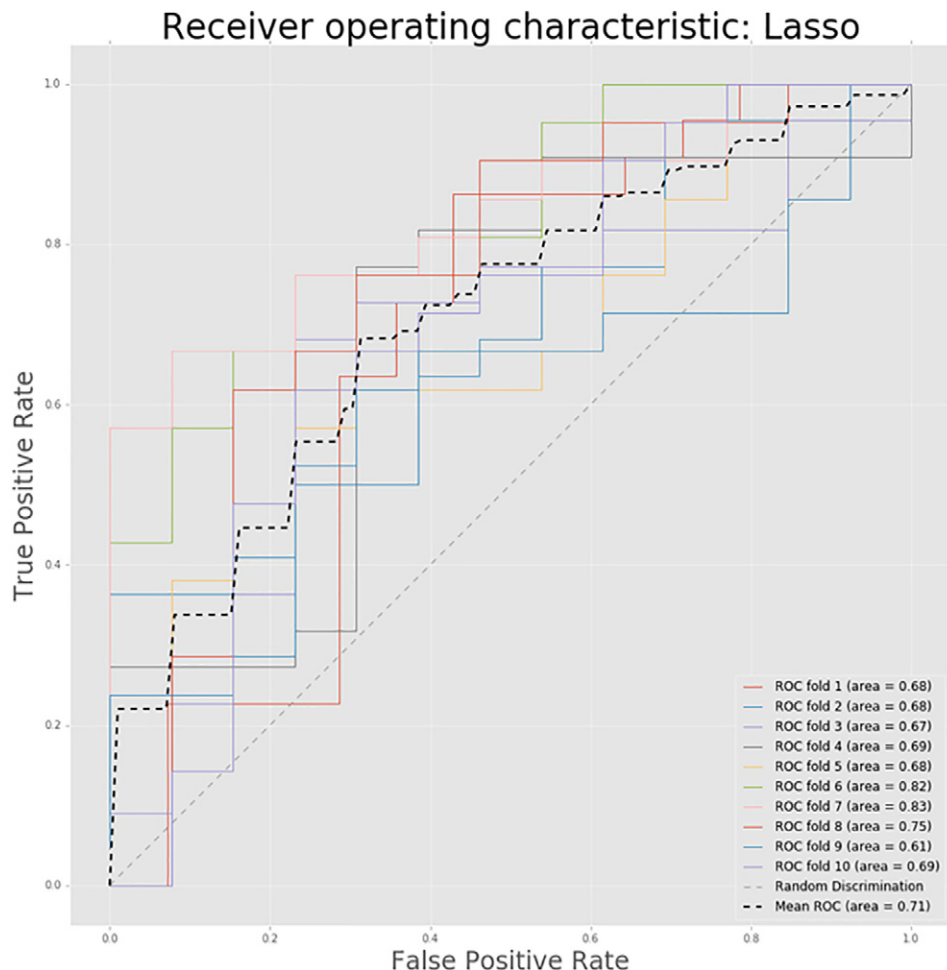


Fig. 2. Lasso ROC curve. ROC curves for 10 folds including the average curve.

We also collected data using the Brief Psychiatric Rating Scale [BPRS] [24]; however, we did not include the BPRS score in the analysis since the severity is specifically related to the presentation at the time of the assessment.

Further clinical information regarding familial history of suicidal behaviour, psychosis, and mood disorders was collected using the Family Interview for Genetic Studies [13]. Information on the participants' primary language and religious affiliation were also collected.

2.3. Predictor variable selection

We included 27 predictor variables in our model based on *a priori* evidence for an increased risk of suicide attempt in schizophrenia. The variables selected were associated with immigration and ethnicity, including whether or not they were born in Canada [3,36]; their age at immigration [21]; first language (English or Other) and religion (Religious Affiliation or No Religion) [34]. Furthermore, important clinical risk factors putatively associated with suicide risk in schizophrenia were included such as age [17]; sex [16]; duration of illness [7]; number of hospitalizations [38]; family history of mood disorder [35], psychosis [4], and suicide [31,39], and lifetime history of alcohol, cannabis and drug abuse/dependence [38]. Scores from clinical measures previously associated with suicide attempt were also incorporated including NEO-FFI [27] and CTQ [15]. Since the predictor variables have different value ranges in which some are discrete variables and others are continuous variables they were scaled according to z-score normalization using Python. The selected predictor variables were used in the machine learning algorithms to predict the classification output labels (suicide

attempter or non-attempter). Demographic and clinical data with predictor variable characteristics are summarized in Table 1. We performed a univariate logistic regression analysis for all predictor variables with suicide attempt status as the outcome. All statistical analyses were carried out in R v. 3.3.0.

2.4. Classification algorithms

The clinical sample was split according to the history of suicide attempt. Our goal was to correctly classify individuals in our sample based on this binary classification. We utilized four machine learning algorithms for classification (LASSO and Random Forest, support vector classifier and elastic net) with Python 2.7, using the scikit-learn package, which were run on the Specialized Computing Cluster (SCC) at CAMH.

The LASSO technique implements a standard logistic regression formula but adds a 'penalization' procedure by assigning some predictor variable weights as zero, referred to as regularization. This latter procedure is critical for improving the predictive accuracy of the binary logistic regression model and allows for improved generalizability to different datasets.

The RF model works by creating a multitude of decision trees and providing an output class. RF produces a collection of randomly generated decision tree predictors. Each branching node is equivalent to one of the input variables. RF selects a subset of input variables to construct a tree, whereby each tree decides the most likely class for the given inputs and picks the most frequent class among all the trees. For the RF model, the whole procedure was repeated 20 times to reduce testing error.

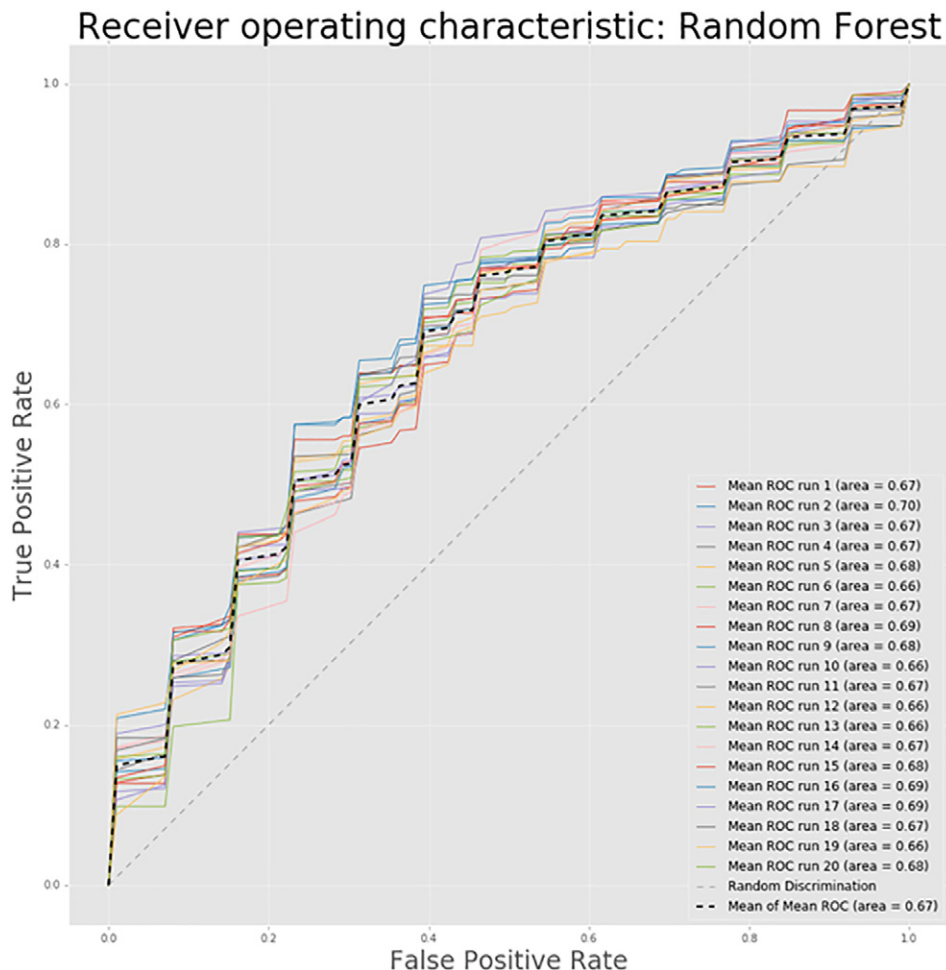


Fig. 3. RFC ROC. Average curve of 20 runs (each run included 10 folds).

Support Vector Classifier (SVC) is a type of support vector machine (SVM). It uses high dimensional hyperplanes to classify and regress given variables. The vector hyperplanes separates the data points into different areas of correlation (with one another). In this case there will be only two areas of correlation: predicted suicide attempters and predicted non-attempters. With the use of a linear kernel, statistical importance weights can be assigned to the clinical predictors used.

Elastic net regularization is a hybrid approach using both L1 penalty of Lasso regression and L2 penalty of Ridge regression. It features a hyper-parameter from the range of 0 to 1 determining how much L2 penalty (at 0) and L1 penalty (at 1) is used.

The overall goal of all four algorithms is to create a model that predicts the classification label based on many input variables.

2.5. Model testing and cross-validation

In a given dataset there may be a class imbalance in which the participants in one class (suicide non-attempters) are more frequent than participants in the other class (suicide attempters) [9]. In our splitting of training and testing data we incorporated a Stratified K-Fold cross-validation procedure. The data was first partitioned into k equally sized folds. The folds were made by preserving the ratio for suicide attempters and non-attempters. Here, we assigned $k = 10$, whereby 10 folds of training and validation were done, such that each fold is used once as the test set while the remaining $k-1$ folds are used for training. In our current classification problem, there are an unequal number of attempters and non-attempters. Our performance summary statistics were calculated as an average across all folds. Average accuracy and area

under the curve (AUC) scores with their standard errors (SE) are generated using values averaged across all folds. A summary of the procedure is illustrated in Fig. 1.

Model comparisons were done using Receiver Operating Characteristic (ROC) curves and their corresponding AUC scores. Significant differences across the four models were tested using the McNemar's test. In order to assess the validity of our models in correctly identifying individuals as suicide attempters or non-attempters we calculated specific model statistics, including prediction accuracy, sensitivity, specificity, positive predictive values (PPV), and negative predictive values (NPV). Sensitivity is referred to as the true positive rate which measures the proportion of positives (suicide attempters) that were correctly identified. Specificity (true negative rate) is a measure of the proportion of negatives (suicide non-attempters) that were correctly identified. Predictive accuracy was calculated as the sum of true positives and true negatives, divided by the total sample. PPV (%) was calculated as the proportion of the truly predicted attempters against all attempters and NPV (%) was calculated as the proportion of the truly predicted non-attempters against all non-attempters.

3. Results

Our final sample consisted of 345 patients with schizophrenia spectrum disorders in which 131 subjects attempted suicide lifetime. All individuals included in the model had no missing data for the included predictor variables. Any missing data was retrieved through medical chart review when possible. The effect of the single predictors on suicide attempt is summarized in Table 1.

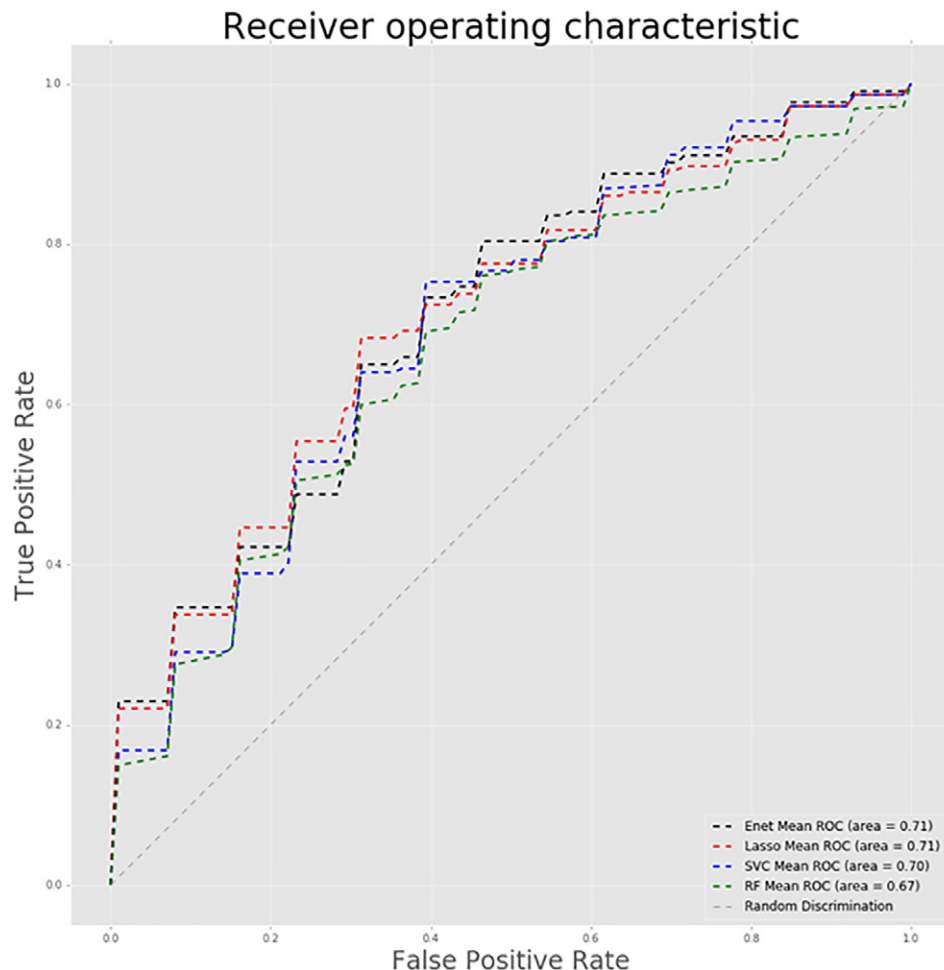


Fig. 4. RFC, lasso, EN and SVC average ROC curves. ROC curves for 10 folds including the average curve.

The LASSO model was able to correctly identify suicide attempters and non-attempters with a cross-validated prediction mean accuracy of 0.67 ± 0.02 . The corresponding ROC curve (Fig. 2) and confusion matrix (Fig. 5) for all stratified folds were used to calculate a cross-validated sensitivity and specificity. The algorithm was able to generate an average sensitivity of 64% and average specificity of 68%. By creating ROC curves for all ten stratified folds, we generated an average AUC of 0.71 ± 0.02 . We also calculated an average PPV of 67% and a NPV of 66% from our model.

Within the LASSO regression model, the top five predictor variables averaged across all folds for the model were duration of illness, number of hospitalizations (>5), CTQ emotional abuse item, lifetime drug abuse/dependence, and the CTQ physical abuse item (Fig. 6).

The RF model was able to correctly identify suicide attempters and non-attempters with a cross-validated prediction mean accuracy of 0.66 ± 0.00 . The corresponding ROC curve for all repetitions (Fig. 3) and confusion matrix (Fig. 5) were used to calculate a cross-validated sensitivity and specificity. The algorithm was able to generate an average sensitivity of 45% and an average specificity of 80%. Using 20-repetition mean of 10-fold ROC curves, we generated an average AUC of 0.67. We also calculated a PPV of 68% and a NPV of 60% from our model.

The top five predictor variables averaged across all folds and repetitions for the RF model were age, CTQ emotional abuse, duration of illness, CTQ total score, and NEO Neuroticism (Fig. 6).

The SVC model was able to correctly identify suicide attempters and non-attempters with a cross-validated prediction mean accuracy of 0.66 ± 0.01 . The corresponding ROC curve (Fig. 4) and confusion matrix (Fig. 5) for all stratified folds were used to calculate a cross-validated sensitivity and specificity. The algorithm was able to generate an average sensitivity of 63% and average specificity of 68%. By creating ROC curves

for all ten stratified folds, we generated an average AUC of 0.70 ± 0.02 . We also calculated an average PPV of 67% and a NPV of 66% from our model.

The Elastic Net model was able to correctly identify suicide attempters and non-attempters with a cross-validated prediction mean accuracy of 0.65 ± 0.02 . The corresponding ROC curve for all stratified folds (Fig. 4) and confusion matrix (Fig. 5) were used to calculate a cross-validated sensitivity and specificity. The algorithm was able to generate an average sensitivity of 65% and an average specificity of 65%. Using 10-fold ROC curves we generated an average AUC of 0.71 ± 0.02 . We also calculated a PPV of 65% and a NPV of 66% from our model.

A comparison of the performance of all LASSO, RF, SVC, and Elastic Net algorithms is given in Table 2. The four models did not differ in pair-comparison as tested using the McNemar's test (Table 3).

4. Discussion

Many studies have attempted to identify clinically relevant risk factors to preemptively identify individuals with schizophrenia who are most likely to attempt suicide. Unfortunately, the strength of these risk factors varies between studies and largely produces false positive effect. Although many risk factors have been put forward, it has remained an elusive goal to synthesize these into a clinically relevant tool to predict patients who are at the greatest risk for suicide attempt, at the individual level. By using machine learning algorithms, this is the first study, to our knowledge, to bring together many potential risk factors to identify the risk for schizophrenia patients attempting suicide. Our algorithms demonstrated a predictive accuracy of 67–65% in correctly identifying suicide attempters and non-attempters, which is

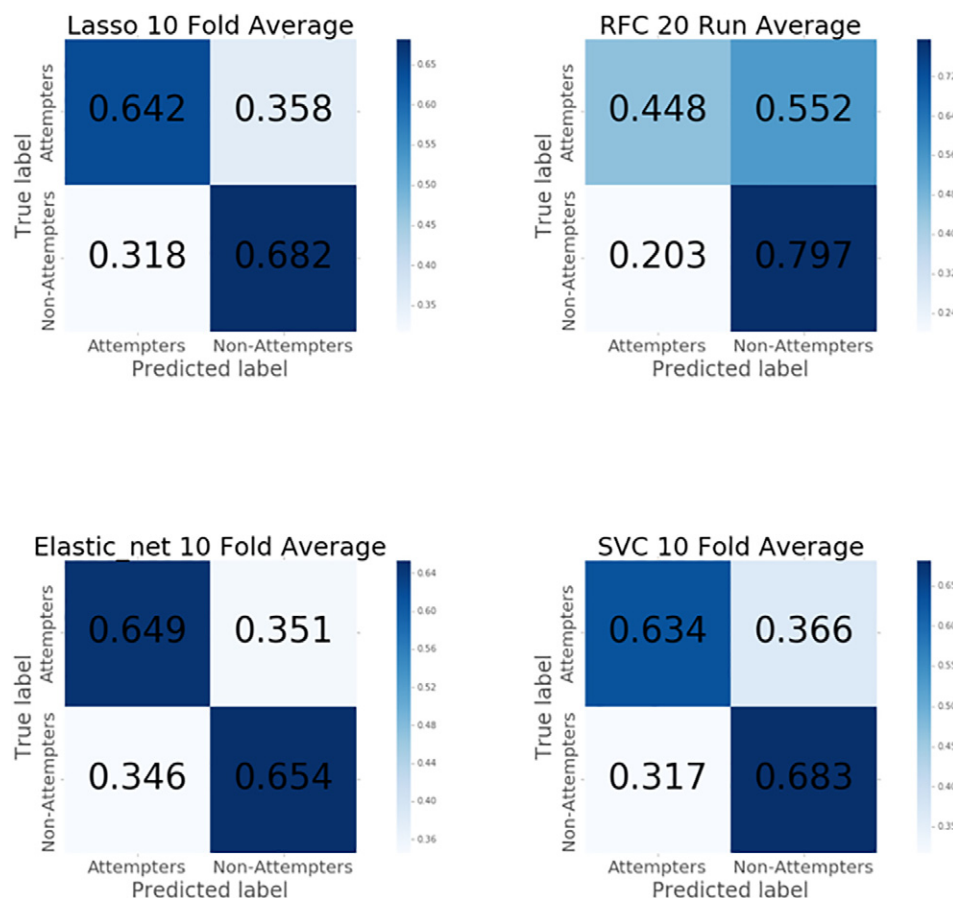


Fig. 5. Confusion matrix. Confusion matrix for all four models.

greater than what would have been determined by chance alone. All models performed equally well, in terms of the aforementioned prediction accuracy with AUC scores between 0.71 and 0.67. Our model statistics were validated using a stratified 10-fold cross-validation approach. This prevented an inherent bias in our training and testing data to ensure that all models were able to train on data which maintained the original distribution of suicide attempters and non-attempters. Our machine learning algorithms were ultimately able to determine the probability of distinguishing an individual as a suicide attempter or non-attempter with confidence using the predictor variables we included.

The predictor variables included in our model were based on *a priori* knowledge of risk factors for suicide attempt in schizophrenia from the scientific literature. Current classifications in psychiatry rely on the observation of individual symptoms to reliably make a clinical diagnosis. This is particularly difficult for many psychiatric phenotypes, especially for suicidality. Suicidal behaviour ranges on a continuum and varies in its lethality and risk. Here, we focused on classifying individuals with schizophrenia spectrum disorders as either suicide attempters or non-attempters. Particularly in schizophrenia, suicide attempts may occur without any obvious warnings or clear signs; making it difficult to predict [38]. Unique to schizophrenia spectrum disorders, patients may be driven to attempting suicide due to delusional thoughts or hallucinations. Therefore, suicide attempting is difficult to categorize under these circumstances. Failing objective methods of determining intentionality, some regard suicide attempts as “parasuicide” or “deliberate” or “non-fatal” self-injury instead, where there is believed to be intention to die at least greater than zero. Furthermore, suicide attempts vary widely in their lethality and method choices. Classifying all suicide attempters as a single diagnostic label may therefore mask the subtleties in risk factors that lead to different suicidal behaviours. Because

the diagnostic reliability of a suicide attempt as a phenotype may lack some reliability and under-represents inter-individual differences, using suicide attempt status as a classification label may prove to be ineffective in developing robust machine learning predictive models [12]. Having a clear, distinctive classification label is one way in which these models will improve in their reliability and generalization to a larger population, which may require an improved system for subtyping behaviour we currently term loosely as suicide attempts.

Although our study is arguably the first to attempt to utilize machine learning algorithms to distinguish suicide attempters from non-attempters in a sample of schizophrenia patients, other studies have attempted to do the same in other diagnostic domains. In one study, a signature for suicide attempts in mood disorders was derived using machine learning models [26]. The authors found in their most predictive model, using a relevance vector machine (RVM) algorithm, an AUC of 0.77 with 72% accuracy. Compared to our LASSO model which was our most predictive with an AUC of 0.71 and 67% accuracy our algorithms performed quite consistently for the given phenotype. Despite using a much smaller sample size, different machine learning techniques, different predictor variables, and investigating suicide attempt in schizophrenia as opposed to mood disorders, our results do not substantially differ from the previous reports. This leaves the open question as to the effectiveness of the same models in predicting suicide attempt risk across many diagnostic groups.

Using clinical and demographic predictor variables alone may not be sufficient in predicting the risk for suicide. Future studies may benefit from incorporating a biological hypothesis for suicide attempt in which candidate genes [8] or imaging data [20,23] may be included in these models. In studies that have attempted to identify risk factors associated with suicide in schizophrenia, many of the same risk clinical

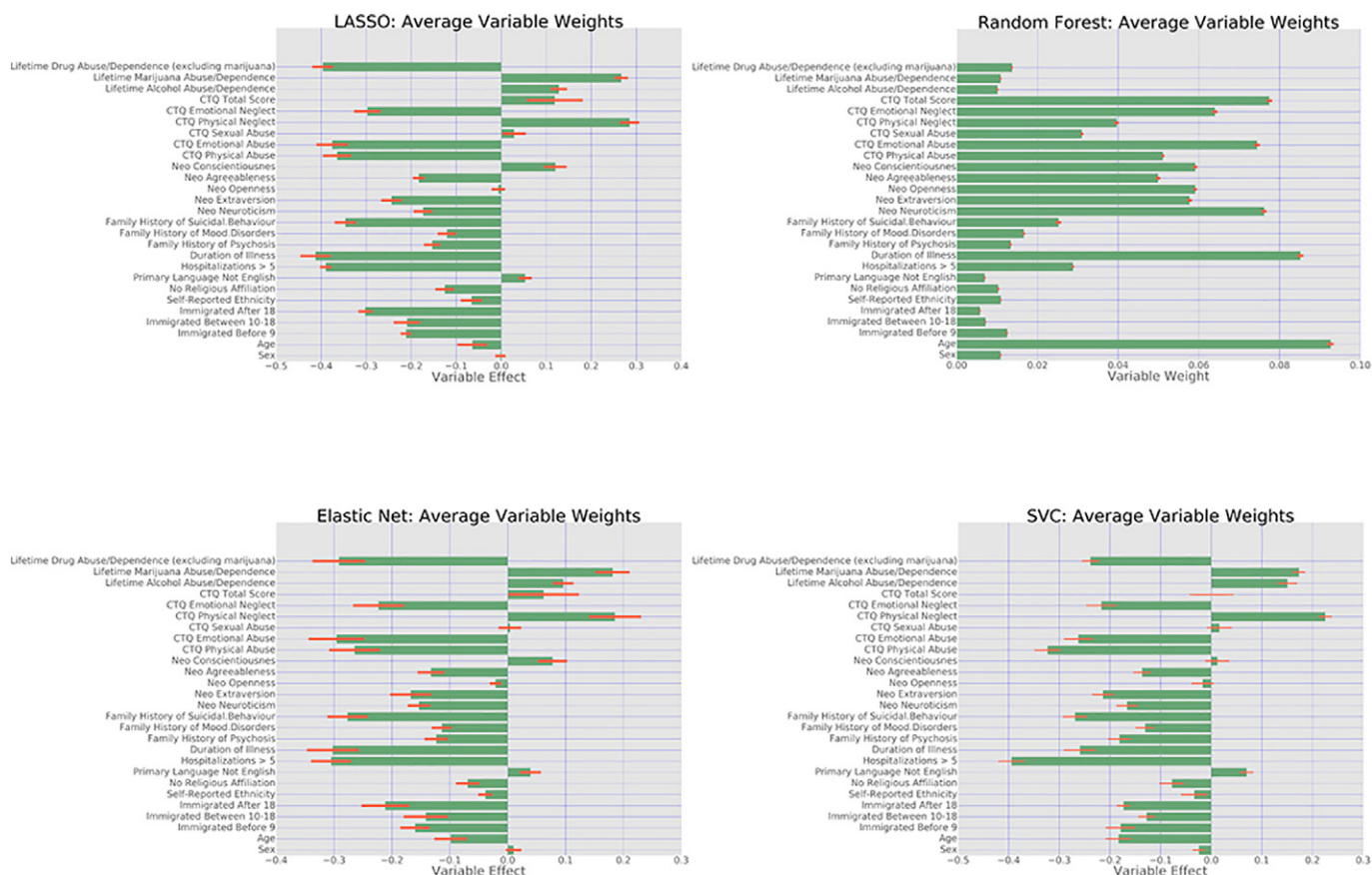


Fig. 6. Average variable weights for four models. Comparison of the effect size for each model; data shown as mean and SEM of 10 folds (and 20 repetitions for random forest model).

Table 2

Comparison of algorithm performance in classifying individual patients as either suicide attempters or non-attempters. Values are provided as means and standard errors.

Algorithm	LASSO	Random forest	Support vector classifier	Elastic net
Area under curve (AUC)	0.71 ± 0.02	0.67 ± 0.00	0.70 ± 0.02	0.71 ± 0.02
Accuracy	0.67 ± 0.02	0.66 ± 0.00	0.66 ± 0.01	0.65 ± 0.02
Sensitivity	0.64 ± 0.03	0.45 ± 0.006	0.63 ± 0.03	0.65 ± 0.04
Specificity	0.68 ± 0.03	0.80 ± 0.003	0.68 ± 0.03	0.65 ± 0.03
PPV	0.67 ± 0.02	0.68 ± 0.006	0.67 ± 0.02	0.65 ± 0.02
NPV	0.66 ± 0.02	0.60 ± 0.003	0.66 ± 0.02	0.66 ± 0.02

factors have been repeatedly suggested [38]. However, no studies to our knowledge, have attempted to perform a large-scale analysis of risk factors for suicide attempt in schizophrenia using clinical, genetic, and imaging data to define a clear risk for suicide. Future studies of this kind may therefore benefit from the incorporation of risk factors clearly identified by these multimodal studies that combine data across all dimensions.

To our knowledge, as the first study to use machine learning algorithms to classify suicide attempters and non-attempters in schizophrenia, our models are only reasonably predictive. Notably, compared to other classification studies for suicide [14,19,26] our models perform nearly as effectively.

The limitations of our study, which may have prevented higher AUC and accuracy scores, include our relatively small sample size and limited number of predictor variables. Also, the small sample size may have affected the comparison test among different models, showing that the four models tested do not differ. Therefore, Machine learning models for complex classification problems such as this mainly benefit from larger sample sizes with many predictor variables in order to better train the models. Furthermore, we chose to incorporate socio-cultural risk factors for suicide to diversify our predictor variables. Unfortunately, it appeared that these variables had little predictive value in our models. Nevertheless, our models' strength lies in the robustness of our classifier label in which individual clinical assessments were conducted with both interview and self-report questionnaires along with medical chart reviews to assess patients' history of suicide attempt.

Another limitation of our study is that we were able to apply only four machine learning models and this could have prevented us to test more accurate models.

In conclusion, we reported the use of four machine learning models with acceptable accuracy in stratifying a sample of patients with schizophrenia as either suicide attempters or non-attempters using socio-cultural, clinical, and demographic predictor variables. Our findings suggest that these machine learning models have some clinical utility in classifying schizophrenia patients according to their risk for suicide. Indeed, machine learning models with binary classification algorithms may one day complement and improve the diagnostic reliability of many psychiatric phenotypes. By utilizing large scale clinical data and advanced probabilistic algorithms we may be able to develop highly predictive models that will support clinical decisions related to suicide risk. Doing so can improve therapeutic interventions for patients with schizophrenia by accurately and swiftly identifying patients who may be at the highest risk for a future suicide attempt. Nevertheless, improvements to our model are contingent on a multimodal approach to integrating genetic, neuroimaging, and more predictive clinical

variables to accurately predict suicide risk in a cohort of patients with schizophrenia.

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Conflict of interest

Authors declare no conflict of interest.

Ethical standards

The authors assert that all procedures contributing to this work comply with the ethical standards of the institutional review board of the Centre for Addiction and Mental Health and with the Helsinki Declaration of 1975, as revised in 2008. Participation was voluntary and anonymous; informed consent was obtained from all participants. Compensation was offered to participants.

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Table 3

McNemar test comparison across models.

Algorithm	LASSO	Random forest	Elastic net	Support vector classifier
LASSO	–	0.566	0.302	1.000
Random forest	0.329	–	0.235	0.550
Elastic net	1.067	1.408	–	0.332
Support vector classifier	0.000	0.357	0.941	–

Above the diagonal: p-values; below the diagonal: chi-square values.

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